

Alpilla 600 MW Combined Cycle Power Plant Plant Design Basis Report

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Rev	Purpose	Date	Description	Prepared	Checked	Approved
A	IFR	2026-11-16	First issue for Owner approval (Basic Design Package)	EPCE	EPC process lead	EPC engineering manager

1. Purpose and scope

This report fixes the design basis of Alpilla 600 MW Combined Cycle Power Plant: the site, climatic, fuel, grid, water and environmental conditions, the performance and operating requirements and the guarantees against which the plant is designed. All values are taken from the project database; each value states its status (assumption, preliminary, confirmed) and its basis (Owner document, requirement, reference or decision). Discipline design criteria (process, mechanical, electrical, I&C, civil / structural, HVAC, HSE) are derived from this report.

2. Plant configuration

1 x 1 multi-shaft combined cycle: one gas turbine with generator, one three-pressure reheat heat recovery steam generator with SCR, one reheat condensing steam turbine with its own generator and a seawater-cooled surface condenser; both generators connected through generator step-up transformers to the 380 kV GIS switchyard. Main equipment (register):

Tag	Description	System	Supply
00XJA10	Emergency diesel generator 1,250 kVA	00XJA	IEPC
10BAT10	GT step-up transformer 560 MVA 400/21 kV	10BAT	IEC
10HAA10	Heat recovery steam generator, 3-pressure reheat	10HAA	IEC
10MBA10	Gas turbine	10MBA	IEC
10MKA10	GT generator 560 MVA, H2-cooled	10MKA	IEC
20BAT10	ST step-up transformer 270 MVA 400/15.75 kV	20BAT	IEC
20MAA10	Steam turbine, reheat condensing, 2 casings	20MAA	IEC
20MKA10	ST generator 270 MVA, TEWAC	20MKA	IEC

92 plant systems (KKS, ALP-EPC-00000-GE-PRC-0001), by category:

Category	Systems
auxiliary mechanical	00QCA Chemical dosing, 00QEA Compressed air, 00QHA Auxiliary boiler, 00QJA Nitrogen supply, 00QJB Hydrogen and CO2 supply, 00QUA Sampling and SWAS, 10HSB Ammonia storage and supply
civil structural	00GQA Storm water drainage
common	00BXA Earthing and lightning protection
condensate feedwater	10LAB Feedwater piping, 10LAC Boiler feed pumps, 10LAE HP attemperation spray water, 10LAF Reheat attemperation spray water, 20LCA Main condensate piping, 20LCB Condensate extraction pumps
cooling	00PAA CW intake, screens and outfall, 00PAB CW piping, 00PAC CW pumps, 00PAH Condenser tube cleaning, 00PBA Electro-chlorination, 00PGA Closed cooling water
electrical hv	00ACA 380 kV GIS switchyard, 10BAA GT generator busduct, 10BAC GT generator circuit breaker, 10BAT GT generator step-up transformer, 20BAA ST generator busduct, 20BAC ST generator circuit breaker, 20BAT ST generator step-up transformer
electrical mv lv	00BBA MV switchgear 10.5 kV, 00BFA LV switchgear 0.4 kV, 00BFT LV transformers, 00BLA Lighting and small power, 00BMA Emergency LV switchgear, 00BRA UPS, 00BTA Batteries and DC 220 V, 00XJA Emergency diesel generator, 10BBT GT unit auxiliary transformer, 20BBT ST unit auxiliary transformer
fire protection	00SGA Fire water system, 00SGC Deluge, spray and foam systems
fuel gas	00EGA LDO unloading, 00EGB LDO storage, 00EGC LDO forwarding and treatment, 00EKA Gas receiving, filtration and fiscal metering station, 00EKC Fuel gas distribution to GT and auxiliary boiler
gas turbine	10MBA Gas turbine, 10MBL GT inlet air system, 10MBN GT liquid fuel and water injection, 10MBP GT fuel gas system incl. performance heater, 10MBU GT compressor washing, 10MBV GT lube oil system
generator	10MKA GT generator, 10MKC GT excitation and static frequency converter, 10MKG GT generator gas system (H2/CO2), 20MKA ST generator, 20MKC ST excitation
hrsg	10HAA HRSG pressure system, 10HNA HRSG flue gas ducting, 10HNE Stack 65 m with CEMS platform and silencer, 10HSA SCR catalyst and ammonia injection
hvac	00SAA HVAC
i and c	00CJA Plant DCS, 00CJG Safety instrumented systems (BOP), 00CKA Historian, performance monitoring and operator training simulator, 00CQA Continuous emission monitoring, 00CYA Telecom, CCTV, access control, security, 00SGY Fire detection and alarm, 10MBY GT control and protection (TCS), 20MAY ST control and protection

Category	Systems
other	00SMA Cranes and hoists
steam turbine	20MAA HP turbine, 20MAB IP/LP turbine, 20MAJ Condenser vacuum system, 20MAV ST lube oil system, 20MAW Gland steam system, 20MAX ST control fluid system
water steam cycle	00LBG Auxiliary steam, 10HAN HRSG blowdown and drains (EPC tanks and headers), 10LBA Main steam piping, 10LBB Hot reheat piping, 10LBC Cold reheat piping, 10LBD LP steam piping, 20MAG Condensing system, 20MAN Turbine bypass stations and piping
water treatment	00GBA Seawater reverse osmosis, 00GCA Demineralisation, 00GHA Demineralised water storage and distribution, 00GKA Potable water, 00GMA Process drains and waste water collection, 00GNA Waste water treatment, 00GUA Sanitary waste water treatment, 00LDA Condensate polishing

3. Site data

Site location, levels, loads and soil conditions for the civil and structural design; site class and seismic hazard per TBDY 2018.

Parameter	Value	Condition	Status	Basis
Atmospheric corrosivity category	C5 (ISO 12944-2), durability H -		confirmed	ER-02.08 Corrosive coastal atmosphere
Basic wind speed	30 m/s	10 min mean, 10 m, 50-year return	confirmed	ER-02.05 Wind, snow and rain
Design rainfall intensity	110 mm/h	10 min, 25-year return	confirmed	ER-02.05 Wind, snow and rain
Distance to heavy-lift port	18 km by sea	400 t quay crane	confirmed	SRC-OWNR-0005 Site, existing infrastructure and tie-in points
Ground snow load	0.75 kN/m2		confirmed	ER-02.05 Wind, snow and rain
Groundwater level	13.5 m a.s.l.	brackish, chloride 3,000-8,000 mg/l	confirmed	SRC-OWNR-0008 Geotechnical baseline summary
Heavy transport road limit	120 t gross	two bridges on the state road	confirmed	SRC-OWNR-0005 Site, existing infrastructure and tie-in points, ER-02.14 Access and transport
Height datum	TUDKA-99 orthometric heights (m a.s.l.)		confirmed	SRC-EPCE-0011 Topographic survey point listing - plot, temporary area TP-A1 and coastal strip
Plant grid origin in UTM zone 35N (E 0 / N 0)	E 541,250.000 m / N 4,532,480.000 m	TUREF (ITRF96), south-west plot corner	confirmed	SRC-EPCE-0011 Topographic survey point listing - plot, temporary area TP-A1 and coastal strip
Platform level acceptance tolerance	0.15 m	Owner-levelled platform +15.00 m	confirmed	DEC-OWNR-0002 Acceptance tolerance of the Owner-levelled platform
Plot dimensions	400 x 300 m	12.0 ha; usable about 10.6 ha	confirmed	SRC-OWNR-0005 Site, existing infrastructure and tie-in points, ER-02.01 Site location and area
Rotation plant grid to UTM grid	-0.3281 deg	plant north = true north; grid convergence at site	confirmed	SRC-EPCE-0011 Topographic survey point listing - plot, temporary area TP-A1 and coastal strip
Scale factor plant grid to UTM	0.999612 -	combined UTM point scale and height factor	confirmed	SRC-EPCE-0011 Topographic survey point listing - plot, temporary area TP-A1 and coastal strip
Seismic design PGA (DD-2, 475-year)	0.45 g	TBDY 2018, site-specific hazard study	confirmed	ER-02.06 Seismic design
Seismic site class	ZD (Vs30 = 240 m/s) -	TBDY 2018	confirmed	SRC-OWNR-0008 Geotechnical baseline summary
Site grade level	15 m a.s.l.		confirmed	ER-02.01 Site location and area

4. Ambient and climatic conditions

Design ambient conditions, Site Reference Conditions (SRC) for performance, salt-laden marine atmosphere and lightning data.

Parameter	Value	Condition	Status	Basis
Ambient temperature - maximum for full capability	40 degC		confirmed	ER-02.02 Ambient design range
Ambient temperature - minimum for full capability	-8 degC		confirmed	ER-02.02 Ambient design range
Ambient temperature - survival range	-15 to +45 degC		confirmed	ER-02.02 Ambient design range
Annual mean temperature (measured)	14.9 degC	2015-2024 site mast	confirmed	SRC-OWNR-0001 Meteorological and ambient air data

Parameter	Value	Condition	Status	Basis
Chloride deposition maximum	420 mg/m2/day	ISO 9225, category S3	confirmed	SRC-OWNR-0001 Meteorological and ambient air data, ER-02.10 Airborne salt and dust
Ground flash density	2.4 flashes/km2/year	26 thunderstorm days/year	confirmed	SRC-OWNR-0001 Meteorological and ambient air data, ER-02.12 Lightning
Performance test ambient range	5 to 30 degC		confirmed	SRC-OWNR-0001 Meteorological and ambient air data, ER-02.11 Design ambient points
PM10 maximum (event)	260 ug/m3	Saharan dust episodes	confirmed	SRC-OWNR-0001 Meteorological and ambient air data, ER-02.10 Airborne salt and dust
Record maximum / minimum temperature	39.6 / -11.4 degC	1960-2024	confirmed	SRC-OWNR-0001 Meteorological and ambient air data
Sea-salt aerosol annual mean	14 ug/m3	as NaCl	confirmed	SRC-OWNR-0001 Meteorological and ambient air data
Sea-salt aerosol maximum (event)	165 ug/m3	as NaCl, lodos storms	confirmed	SRC-OWNR-0001 Meteorological and ambient air data, ER-02.10 Airborne salt and dust
SRC ambient dry-bulb temperature	15 degC	Site Reference Conditions	confirmed	ER-02.03 Site Reference Conditions (SRC)
SRC barometric pressure	1,011.5 mbar	Site Reference Conditions	confirmed	ER-02.03 Site Reference Conditions (SRC)
SRC relative humidity	70 %	Site Reference Conditions	confirmed	ER-02.03 Site Reference Conditions (SRC)
Summer design dry bulb (0.4 %)	32.8 degC	coincident RH 48 %	confirmed	SRC-OWNR-0001 Meteorological and ambient air data, ER-02.11 Design ambient points
Winter design dry bulb (99.6 %)	-4.8 degC	coincident RH 88 %	confirmed	SRC-OWNR-0001 Meteorological and ambient air data, ER-02.11 Design ambient points

5. Fuels

Natural gas (main fuel) at the terminal point.

Parameter	Value	Condition	Status	Basis
Booked gas capacity (firm)	120,000 Sm3/h	firm from the first gas date (TQ-040), 130,000 interruptible	confirmed	SRC-OWNR-0003 Natural gas quality and supply conditions
Exceptional minimum supply pressure	40 barg	max 72 h/year, 24 h notice	confirmed	SRC-OWNR-0003 Natural gas quality and supply conditions
Fuel gas superheat above dew points at GT skid	28 K	minimum	confirmed	ER-03.06 Natural gas contaminant limits
Fuel LHV range	44.2 to 50.0 MJ/kg		confirmed	ER-03.02 Natural gas specification
Gas supply pressure at terminal point	45 to 70 barg		confirmed	ER-03.03 Gas supply conditions at terminal point
Gas supply temperature at terminal point	5 to 25 degC		confirmed	ER-03.03 Gas supply conditions at terminal point
H2S + COS limit	5 mg/Sm3		confirmed	SRC-OWNR-0003 Natural gas quality and supply conditions, ER-03.06 Natural gas contaminant limits
Hydrocarbon dew point	-2 degC (1-70 barg)	maximum	confirmed	SRC-OWNR-0003 Natural gas quality and supply conditions
Hydrogen admixture capability	5 vol %	without modification; 10 vol % by future OEM retrofit (provisions included)	confirmed	ER-03.02 Natural gas specification
Mercury limit	1 ug/Sm3	measured < 0.01	confirmed	SRC-OWNR-0003 Natural gas quality and supply conditions, ER-03.06 Natural gas contaminant limits
Methane content range	85.0 to 98.5 mol %		confirmed	SRC-OWNR-0003 Natural gas quality and supply conditions, ER-03.02 Natural gas specification
Na + K, Pb, V in gas (network data)	< 0.005 (detection limit) mg/Sm3	not guaranteed by network operator	confirmed	SRC-OWNR-0003 Natural gas quality and supply conditions, ER-03.06 Natural gas contaminant limits
Reference fuel LHV	47.48 MJ/kg	reference composition, ISO 6976, combustion reference 25 degC	confirmed	ER-03.02 Natural gas specification

Parameter	Value	Condition	Status	Basis
Total sulphur limit	30 mg S/Sm3	measured typical 4, max 17	confirmed	SRC-OWNR-0003 Natural gas quality and supply conditions, ER-03.06 Natural gas contaminant limits
Water dew point	-8 degC at 70 barg	maximum	confirmed	SRC-OWNR-0003 Natural gas quality and supply conditions
Wobbe index range	47 to 52 MJ/Nm3		confirmed	ER-03.02 Natural gas specification

Light distillate oil (back-up fuel).

Parameter	Value	Condition	Status	Basis
LDO annual operating limit	500 h/year	back-up fuel status	confirmed	SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO), SRC-OWNR-0007 Positive EIA decision - binding conditions
LDO autonomy at base load	72 h	two tanks of 50 %	confirmed	SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO), ER-03.08 LDO unloading and storage
LDO density at 15 degC	820 to 845 (840 for sizing) kg/m3		confirmed	REF-CONS-0096 TS EN 590, SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO)
LDO LHV	42.9 MJ/kg	range 42.6-43.2	confirmed	SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO), ER-03.07 Light diesel oil specification and burner limits
LDO Na + K at GT inlet	0.5 mg/kg	maximum; as delivered up to 1.0 (history 1.8)	confirmed	SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO), ER-03.07 Light diesel oil specification and burner limits
LDO resupply capacity	25 trucks/day	30-36 m3 each	confirmed	SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO)
LDO sulphur	10 mg/kg	maximum	confirmed	SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO), ER-03.07 Light diesel oil specification and burner limits
LDO vanadium / lead / calcium at GT inlet	0.5 / 1.0 / 2.0 mg/kg	maximum	confirmed	SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO), ER-03.07 Light diesel oil specification and burner limits

6. Grid connection

Connection to the 380 kV transmission grid (TSO).

Parameter	Value	Condition	Status	Basis
380 kV voltage range	360 to 410 kV	342-420 kV for up to 30 min	confirmed	SRC-OWNR-0006 Grid connection data
Connection capacity	640 MW export	40 MW import	confirmed	SRC-OWNR-0006 Grid connection data
Generator power factor range	0.85 lagging to 0.95 leading -	at rated active power	confirmed	ER-04.08 Reactive capability
Grid connection voltage	380 kV	nominal	confirmed	ER-11.01 Grid connection
Grid frequency	50 Hz	operating range 47.5 to 52.5 Hz per Grid Code	confirmed	ER-04.07 Frequency and voltage support
Short-circuit level at 380 kV (2029)	12 (min) to 40 (max) kA	future 50 kA (2035)	confirmed	SRC-OWNR-0006 Grid connection data
Switchyard short-circuit rating	50 kA		confirmed	ER-11.01 Grid connection

7. Performance and operation

Performance requirements at SRC.

Parameter	Value	Condition	Status	Basis
Auxiliary power consumption	2.2 % of gross	maximum, base load	confirmed	ER-04.10 Auxiliary power
Load ramp rate	35 MW/min	minimum	confirmed	ER-04.05 Load ramp rate
Minimum environmental load	40 % net output	maximum	confirmed	ER-04.06 Minimum environmental load
Net electrical output (guaranteed)	600 MW	SRC, 380 kV terminal point	confirmed	ER-04.01 Net electrical output
Net heat rate LHV (guaranteed)	5,860 kJ/kWh	SRC	confirmed	ER-04.02 Net heat rate
Net heat rate LHV at 60 % load (guaranteed)	6,250 kJ/kWh	SRC	confirmed	ER-04.03 Part-load efficiency

Operating regime and design life.

Parameter	Value	Condition	Status	Basis
Cold start time to base load	180 min	maximum	confirmed	ER-04.04 Start-up times
Design life	30 years		confirmed	ER-01.03 Design life and operating regime
Hot start time to base load	45 min	maximum	confirmed	ER-04.04 Start-up times
Operating hours per year	8,000 EOH/year		confirmed	ER-01.03 Design life and operating regime
Starts per year	250 starts/year	50 cold, 100 warm, 100 hot	confirmed	ER-01.03 Design life and operating regime
Warm start time to base load	90 min	maximum	confirmed	ER-04.04 Start-up times

8. Cooling water and water supply

Once-through seawater cooling; seawater quality at the intake; condenser design.

Parameter	Value	Condition	Status	Basis
Condenser cleanliness factor (design)	0.85 -	HEI	confirmed	ER-07.07 Condenser design standard
Condenser cooling water temperature rise	7 K	maximum, base load	confirmed	ER-07.04 Cooling water temperature rise
Condenser pressure at SRC (maximum)	36 mbar(a)	base load, seawater 16.0 degC	confirmed	ER-07.08 Condenser pressure
Condenser tube velocity	1.8 to 2.2 m/s	titanium	confirmed	ER-07.07 Condenser design standard
Jellyfish load design	20 kg/m3	blooms June-September	confirmed	SRC-OWNR-0002 Seawater quality and marine conditions, ER-08.07 Jellyfish and mucilage
Pycnocline depth	20 to 25 m		confirmed	SRC-OWNR-0002 Seawater quality and marine conditions
Residual chlorine at outfall	0.1 mg/l	maximum	confirmed	ER-08.04 Biofouling control
Seawater boron	2.4 to 3.4 mg/l		confirmed	SRC-OWNR-0002 Seawater quality and marine conditions
Seawater chloride - maximum (upper layer)	14,500 mg/l		confirmed	SRC-OWNR-0002 Seawater quality and marine conditions
Seawater salinity range (upper layer)	21.8 to 26.1 PSU	lower layer 38.2 PSU below pycnocline	confirmed	SRC-OWNR-0002 Seawater quality and marine conditions
Seawater temperature at intake - annual mean (measured)	15.4 degC	12 m depth, station M2	confirmed	SRC-OWNR-0002 Seawater quality and marine conditions
Seawater temperature at intake - maximum	27 degC	August	confirmed	SRC-OWNR-0002 Seawater quality and marine conditions
Seawater temperature at intake - minimum	6.9 degC	February	confirmed	SRC-OWNR-0002 Seawater quality and marine conditions
Seawater temperature range at intake	6 to 27 degC		confirmed	ER-02.04 Seawater conditions
Seawater TSS maximum (storm)	140 mg/l		confirmed	SRC-OWNR-0002 Seawater quality and marine conditions
SRC seawater temperature	16 degC	Site Reference Conditions	confirmed	ER-02.03 Site Reference Conditions (SRC)
Thermal plume excess at 100 m mixing zone	3 K	maximum	confirmed	ER-08.05 Thermal plume
Wave height Hs 50-year	4.4 m	Tp 8.5 s, SW	confirmed	SRC-OWNR-0002 Seawater quality and marine conditions

9. Emissions and noise

Stack emission limits (15 % O2, dry) and stack height.

Parameter	Value	Condition	Status	Basis
CO	30 mg/Nm3	dry, 15 % O2, 50-100 % GT load	confirmed	ER-14.02 Air emissions
Dust on LDO	5 mg/Nm3	dry, 15 % O2	confirmed	SRC-OWNR-0007 Positive EIA decision - binding conditions, ER-14.02 Air emissions
NH3 slip	5 mg/Nm3	dry, 15 % O2	confirmed	ER-14.02 Air emissions
NOx (as NO2)	30 mg/Nm3	dry, 15 % O2, 50-100 % GT load	confirmed	ER-14.02 Air emissions
NOx on LDO	50 mg/Nm3	dry, 15 % O2	confirmed	SRC-OWNR-0007 Positive EIA decision - binding conditions, ER-14.02 Air emissions
SO2 on LDO	10 mg/Nm3	dry, 15 % O2	confirmed	SRC-OWNR-0007 Positive EIA decision - binding conditions, ER-14.02 Air emissions
Stack height	65 m	EIA decision (approved dispersion study)	confirmed	ER-06.06 Stack, SRC-OWNR-0007 Positive EIA decision - binding conditions

Noise limits.

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Parameter	Value	Condition	Status	Basis
Near-field noise	85 dB(A)	1 m from equipment	confirmed	ER-14.03 Near-field noise
Noise at nearest receptor (night)	45 dB(A)	L _{Aeq} night	confirmed	ER-14.04 Environmental noise

10. Guarantees

Owner guarantees (PG, contract ALP-EPC-001) and the power island guarantees of Imaginary Electric (IG, ALP-CA-001) that cover them back-to-back.

Id	Guarantee	Value	Min / max	Conditions
IG-01	Power island gross output at the generator terminals	615 MW	min	Site Reference Conditions SRC-NG-100 (ER-02.03), new and clean, per IEC-ALP-PER-001 section 7
IG-02	Power island gross heat rate (LHV)	5,735 kJ/kWh	max	Site Reference Conditions SRC-NG-100 (ER-02.03), new and clean, per IEC-ALP-PER-001 section 7
IG-03	GT output at the generator terminals	418 MW	min	Site Reference Conditions SRC-NG-100 (ER-02.03), new and clean, per IEC-ALP-PER-001 section 7
IG-04	Condenser pressure at 16.0 degC seawater, 36,050-38,000 m ³ /h (corrected to 36,050 m ³ /h)	35.5 mbar(a)	max	SRC-NG-100, new and clean, ball cleaning in operation and debris filters clean, per IEC-ALP-PER-001 7
IG-05	IEC auxiliary loads (excl. GSU losses)	2,700 kW	max	Site Reference Conditions SRC-NG-100 (ER-02.03), new and clean, per IEC-ALP-PER-001 section 7
IG-06	NO _x at HRSG outlet (15 % O ₂ , dry), 40-100 % GT load, natural gas	30 mg/Nm ³	max	Site Reference Conditions SRC-NG-100 (ER-02.03), new and clean, per IEC-ALP-PER-001 section 7
IG-07	Power island gross heat rate (LHV) at the GT load for 60 % plant net load	6,060 kJ/kWh	max	SRC, sliding pressure, new and clean, per IEC-ALP-PER-001 Rev A section 7
IG-08	GT minimum emissions-compliant load (natural gas)	30 % GT load	max	Natural gas per ER-03.02, NO _x and CO within the stack limits with SCR and oxidation catalyst
IG-09	Power island gross output on LDO at the generator terminals	572 MW	min	SRC, LDO with water/fuel 0.60, new and clean, per IEC-ALP-PER-003 Rev A case SRC-LDO-100
IG-10	Power island gross heat rate (LHV) on LDO	6,020 kJ/kWh	max	SRC, LDO with water/fuel 0.60, new and clean, per IEC-ALP-PER-003 Rev A case SRC-LDO-100
PG-01	Net electrical output at Site Reference Conditions	600 MW	min	Base load, new and clean (corrected for recoverable degradation per agreed curves), Site Reference Conditions (ER-02.03), power factor 0.85 lagging at the 380 kV terminal point, all auxiliaries in normal operation.
PG-02	Net heat rate (LHV) at Site Reference Conditions	5,860 kJ/kWh	max	As PG-01, fuel within the Fuel Specification (ER-03), heat input on LHV basis.
PG-03	Net heat rate (LHV) at 60 % plant load	6,250 kJ/kWh	max	Plant at 60 % of net output, Site Reference Conditions, sliding pressure.
PG-04	NO _x emission (as NO ₂)	30 mg/Nm ³	max	Dry flue gas, 15 % O ₂ , 50 % to 100 % GT load, all ambient conditions within the design range.
PG-05	CO emission	30 mg/Nm ³	max	Dry flue gas, 15 % O ₂ , 50 % to 100 % GT load.
PG-06	Ammonia slip from the SCR	5 mg/Nm ³	max	Dry flue gas, 15 % O ₂ , 50 % to 100 % GT load, catalyst new.
PG-07	Near-field noise	85 dB(A)	max	1 m from equipment surfaces / enclosures, 1.5 m height, all operating modes except emergency.
PG-08	Noise at nearest sensitive receptor (night)	45 dB(A)	max	L _{Aeq} at the nearest residential receptor defined in the EIA, plant at base load, night period.
PG-09	Hot start time to base load	45 min	max	Shutdown shorter than 8 h, GT ignition to plant base load, emissions compliant from minimum environmental load.
PG-10	Load ramp rate in combined cycle	35 MW/min	min	Between minimum environmental load and base load, emissions compliant.
PG-11	Minimum environmental load	40 % of plant net output	max	Lowest load with PG-04/PG-05 compliance at Site Reference Conditions.
PG-12	Reliability run availability	98 %	min	30 consecutive days of operation following Owner dispatch, maximum two forced outages, none longer than 48 h.
PG-13	Condenser cooling water temperature rise	7 K	max	Base load, all seawater temperatures within the design range.
PG-14	Thermal plume at edge of mixing zone	3 K	max	Temperature excess over ambient seawater at 100 m from the outfall diffuser, base load, summer conditions.
PG-15	Net electrical output on LDO at Site Reference Conditions	560 MW	min	Base load on the reference LDO (LHV 42.9 MJ/kg, FAME 0 %, water 100 mg/kg), water injection in service, SRC; corrections from the actual test fuel by agreed curves (TQ-023).
PG-16	Net heat rate (LHV) on LDO at Site Reference Conditions	6,150 kJ/kWh	max	Base load on the reference LDO (LHV 42.9 MJ/kg, FAME 0 %, water 100 mg/kg), water injection in service, SRC; corrections from the actual test fuel by agreed curves (TQ-023).
PG-17	NO _x emission on LDO (as NO ₂)	50 mg/Nm ³	max	Dry, 15 % O ₂ , 50 % to 100 % GT load on LDO.
PG-18	Automatic fuel transfer gas to LDO without trip	100 % successful transfers	min	Three transfers at 50 % load and three at base load, triggered by simulated loss of gas pressure.

11. Codes and standards

The applicable codes, standards and Turkish regulations, their editions and their order of precedence are listed in the Codes, Standards and Regulations Register (Appendix A.19, ALP-EPC-00000-GE-LST-0001); design to Eurocodes with Turkish national annexes and TBDY 2018 for structures, IEC for electrical and I&C, ASME / EN for pressure parts, NFPA 850 for fire protection.

12. Open items and assumptions

Parameter	Value	Status	Basis
Fire water design flow and pressure	341 m3/h	preliminary	REF-CONS-0069 NFPA 850, REF-CONS-0070 NFPA 20, DEC-EPCE-0014 Preliminary equipment sizing basis and conventions
Fire water storage	700 m3	preliminary	REF-CONS-0069 NFPA 850, DEC-EPCE-0014 Preliminary equipment sizing basis and conventions

13. Basis documents

Reference	Title
decision:DEC-EPCE-0014	DEC-EPCE-0014 Preliminary equipment sizing basis and conventions
decision:DEC-OWNR-0002	DEC-OWNR-0002 Acceptance tolerance of the Owner-levelled platform
reference:REF-CONS-0069	REF-CONS-0069 NFPA 850
reference:REF-CONS-0070	REF-CONS-0070 NFPA 20
reference:REF-CONS-0096	REF-CONS-0096 TS EN 590
requirement:ER-01.03	ER-01.03 Design life and operating regime
requirement:ER-02.01	ER-02.01 Site location and area
requirement:ER-02.02	ER-02.02 Ambient design range
requirement:ER-02.03	ER-02.03 Site Reference Conditions (SRC)
requirement:ER-02.04	ER-02.04 Seawater conditions
requirement:ER-02.05	ER-02.05 Wind, snow and rain
requirement:ER-02.06	ER-02.06 Seismic design
requirement:ER-02.08	ER-02.08 Corrosive coastal atmosphere
requirement:ER-02.10	ER-02.10 Airborne salt and dust
requirement:ER-02.11	ER-02.11 Design ambient points
requirement:ER-02.12	ER-02.12 Lightning
requirement:ER-02.14	ER-02.14 Access and transport
requirement:ER-03.02	ER-03.02 Natural gas specification
requirement:ER-03.03	ER-03.03 Gas supply conditions at terminal point
requirement:ER-03.06	ER-03.06 Natural gas contaminant limits
requirement:ER-03.07	ER-03.07 Light diesel oil specification and burner limits
requirement:ER-03.08	ER-03.08 LDO unloading and storage
requirement:ER-04.01	ER-04.01 Net electrical output
requirement:ER-04.02	ER-04.02 Net heat rate
requirement:ER-04.03	ER-04.03 Part-load efficiency
requirement:ER-04.04	ER-04.04 Start-up times
requirement:ER-04.05	ER-04.05 Load ramp rate
requirement:ER-04.06	ER-04.06 Minimum environmental load
requirement:ER-04.07	ER-04.07 Frequency and voltage support
requirement:ER-04.08	ER-04.08 Reactive capability
requirement:ER-04.10	ER-04.10 Auxiliary power
requirement:ER-06.06	ER-06.06 Stack
requirement:ER-07.04	ER-07.04 Cooling water temperature rise
requirement:ER-07.07	ER-07.07 Condenser design standard
requirement:ER-07.08	ER-07.08 Condenser pressure
requirement:ER-08.04	ER-08.04 Biofouling control
requirement:ER-08.05	ER-08.05 Thermal plume
requirement:ER-08.07	ER-08.07 Jellyfish and mucilage
requirement:ER-11.01	ER-11.01 Grid connection
requirement:ER-14.02	ER-14.02 Air emissions
requirement:ER-14.03	ER-14.03 Near-field noise
requirement:ER-14.04	ER-14.04 Environmental noise
source:SRC-EPCE-0011	SRC-EPCE-0011 Topographic survey point listing - plot, temporary area TP-A1 and coastal strip
source:SRC-OWNR-0001	SRC-OWNR-0001 Meteorological and ambient air data
source:SRC-OWNR-0002	SRC-OWNR-0002 Seawater quality and marine conditions
source:SRC-OWNR-0003	SRC-OWNR-0003 Natural gas quality and supply conditions
source:SRC-OWNR-0004	SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO)
source:SRC-OWNR-0005	SRC-OWNR-0005 Site, existing infrastructure and tie-in points
source:SRC-OWNR-0006	SRC-OWNR-0006 Grid connection data
source:SRC-OWNR-0007	SRC-OWNR-0007 Positive EIA decision - binding conditions
source:SRC-OWNR-0008	SRC-OWNR-0008 Geotechnical baseline summary